ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2025 4WM2H Worked Solutions

Nov2025 | 4WM2H | Unit 2

33 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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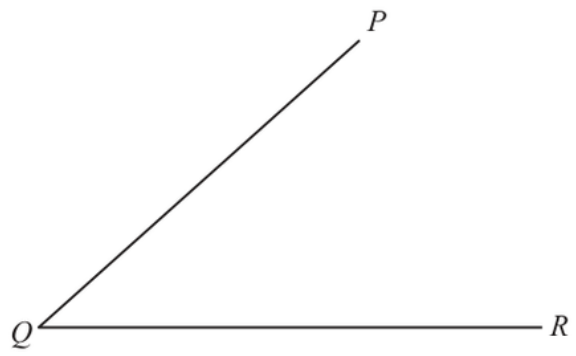
PAST PAPER QUESTION Q#: 1 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

1 Using ruler and compasses only, construct the bisector of angle *PQR*
You must show all your construction lines.



(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 2

TOPIC: **Bearings, Scale Drawing & Constructions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Place the point of the compass at the vertex Q. Draw an arc that intersects both arms QP and QR. Let's call these intersection points A and B.
Step 2: Without changing the compass width, place the compass point on A and draw an arc inside the angle.
Step 3: Place the compass point on B and draw another arc to intersect the previous arc. Let this intersection be point C.
Step 4: Draw a straight line from Q through C. This is the angle bisector.

PAST PAPER QUESTION Q#: 2 Marks: 4

Topic: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

2 The table gives information about the heights, in cm, of 60 flowers in Abby's garden.

Height (h cm)	Frequency
$0 < h \leqslant 10$	17
$10 < h \leqslant 20$	9
$20 < h \leqslant 30$	16
$30 < h \leqslant 40$	14
$40 < h \leqslant 50$	4

Work out an estimate for the mean height of these flowers.

.... cm

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Find the midpoints (x) for each class interval.

$0 < h \leq 10 \implies x = 5$

$10 < h \leq 20 \implies x = 15$

$20 < h \leq 30 \implies x = 25$

$30 < h \leq 40 \implies x = 35$

$40 < h \leq 50 \implies x = 45$

Step 2: Multiply frequency (f) by midpoint (x) to find fx.

Step 3: Calculate the mean.

$$\text{Mean} = \frac{\sum fx}{\sum f} = \frac{1290}{60} = 21.5$$

FINAL ANSWER

21.5 cm

PAST PAPER QUESTION Q#: 3 Marks: 3

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

3 Solve the simultaneous equations

$$5x + y = 11$$
$$3x - y = 9$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION Q#: 3 Marks: 3

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Label the equations:

$$5x + y = 11 \quad — (1)$$
$$3x - y = 9 \quad — (2)$$

Step 1: Add (1) and (2) to eliminate y.

$$(5x + y) + (3x - y) = 11 + 9$$
$$8x = 20$$
$$x = \frac{20}{8} = 2.5$$

Step 2: Substitute $x = 2.5$ into equation (1) to find y.

$$5(2.5) + y = 11$$
$$12.5 + y = 11$$
$$y = 11 - 12.5 = -1.5$$

FINAL ANSWER
$x = 2.5$, $y =$ - 1.5

PAST PAPER QUESTION

Topic: **Percentages**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

Q#: 4 Marks: 4

4 Pete plants a total of 960 trees in a park.
He plants only apple trees, cherry trees and pear trees such that

number of apple trees : number of cherry trees : number of pear trees = 6 : 7 : 3

65% of the cherry trees will grow morello cherries.

Work out the number of trees that will grow morello cherries.

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Percentages**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Find the total number of ratio parts.

$$Total parts = 6 + 7 + 3 = 16 parts$$

Step 2: Find the value of one part.

$$1 \text{ part} = \frac{960}{16} = 60 \text{ trees}$$

Step 3: Calculate the total number of cherry trees.

$$Cherry trees = 7 parts \times 60 = 420 trees$$

Step 4: Calculate 65% of the cherry trees to find morellos.

$$Morello trees = 0.65 \times 420 = 273$$

FINAL ANSWER

273

Always read carefully what the final question asks. A common mistake is stopping at the total number of cherry trees (420) instead of calculating the 65% morello portion.

PAST PAPER QUESTION Q#: 5 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

5 Here are three similar quadrilaterals.

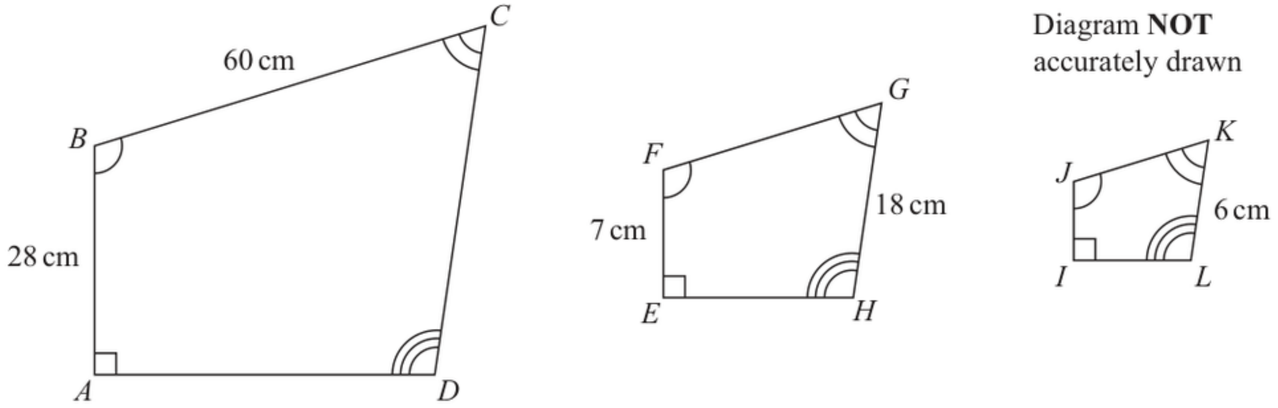

Diagram **NOT** accurately drawn

Work out the length of *JK*

.... cm

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Congruence, Similarity & Geometrical Proof**

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WORKED SOLUTION

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Step 1: Identify corresponding sides from the similar shapes. Looking at the given angles/orientations:
Left side AB corresponds to EF and IJ.
Top side BC corresponds to FG and JK.
Right side CD corresponds to GH and KL.
We are given $CD = 60$ (wait, diagram check: BC is the top side labeled 60, $AB = 28$. On middle shape $EF = 7$, $GH = 18$. On right shape $KL = 6$, JK is unknown). Let's use the explicit lengths:

$$\begin{aligned} AB = 28 &\quad \text{corresponds to} \quad EF = 7 \\ BC = 60 &\quad \text{corresponds to} \quad FG \\ CD &\quad \text{corresponds to} \quad GH = 18 \quad \text{and} \quad KL = 6 \end{aligned}$$

Step 2: Find scale factors. Scale factor from *AB**CD* to *EF**GH* using the left sides:

$$\text{SF}_1 = \frac{EF}{AB} = \frac{7}{28} = \frac{1}{4}$$

So, $FG = BC \times \frac{1}{4} = 60 \times \frac{1}{4} = 15$ cm.
Scale factor from *EF**GH* to *IJ**KL *using the right sides* (GH *and* KL*):

$$\text{SF}_2 = \frac{KL}{GH} = \frac{6}{18} = \frac{1}{3}$$

Step 3: Calculate *JK*. Side JK corresponds to FG.

$$JK = FG \times \text{SF}_2 = 15 \times \frac{1}{3} = 5 \text{ cm}$$

FINAL ANSWER
5 cm

PAST PAPER QUESTION Q#: 6 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

6 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.
On Saturday, Otis sells 87 ice creams.

(a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

....%

(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

(b) Work out the value of the ice cream machine at the end of 3 years.

.... Swiss francs

(3)

(Total for Question 6 is 6 marks)

PAST PAPER SOLUTION Q#: 6 Marks: 6

TOPIC: **Compound Interest & Depreciation**

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WORKED SOLUTION

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Step 1: Calculate the absolute increase.
Increase = 87 - 75 = 12
Step 2: Calculate percentage increase.

$$\% \text{ Increase} = \frac{\text{Increase}}{\text{Original}} \times 100 = \frac{12}{75} \times 100 = 16\%$$

FINAL ANSWER
16%

Step 1: Determine the depreciation multiplier. A 20% decrease means the multiplier is 1 - 0.20 = 0.80.
Step 2: Apply the multiplier for 3 years.

$$FinalValue = 960 \times (0.80)^3$$

$$FinalValue = 960 \times 0.512 = 491.52$$

FINAL ANSWER
491.52 Swiss francs

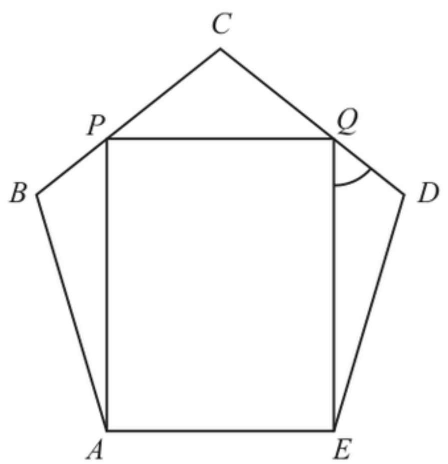
PAST PAPER QUESTION Q#: 7 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

7 *ABCDE* is a regular pentagon and *APQE* is a rectangle.

Diagram **NOT** accurately drawn



BPC and *DQC* are straight lines.

Work out the size of angle *EQD*
Show your working clearly.

.... °

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Interior angle of the regular pentagon.

$$\text{Interior Angle} = \frac{(5-2)\times 180^\circ}{5} = \frac{540^\circ}{5} = 108^\circ$$

So, $\angle AED = 108^\circ$.

Step 2: Find angles within the corner triangle △*DEQ*. Since *APQE *is a rectangle, the angle* ∠AE**Q* = 90°. Therefore, the angle ∠*DEQ*** is:

∠*DE**Q* = ∠*AE**D - ∠AE*Q = 108° - 90° = 18°

Step 3: Analyze △*DEQ*. The problem states *DQC *is a straight line, meaning* Q *lies on the side* DC*. Therefore,* △DE**Q* is formed by *D*, *E*, *Q*. In △*DE**Q*, the angle at vertex* D *is the pentagon's interior angle* ∠CD*E = 108°.

Using angles in a triangle summing to 180°:

∠*EQ**D* = 180° - (∠*DE**Q+∠ED*Q) = 180° - (18°+108°) = 180° - 126° = 54°

FINAL ANSWER

54°

PAST PAPER QUESTION Q#: 8 Marks: 4

TOPIC: **Percentages**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

8 Shop A and shop B have offers for buying the same type of chair.

Shop A	Shop B
25% off the normal price	$\frac{2}{9}$ off the normal price
Sale price 240 euros	Sale price 245 euros

The normal price of the chair in shop A is more than the normal price of the chair in shop B

How much more?
Show your working clearly.

.... euros

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Percentages**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Calculate Normal Price for Shop A. A 25% discount means the sale price is 75% of the normal price.

$$0.75 \times Normal_A = 240$$
$$\text{Normal}_A = \frac{240}{0.75} = 320 \text{ euros}$$

Step 2: Calculate Normal Price for Shop B. A discount of $\frac{2}{9}$ means the sale price is $1 - \frac{2}{9} = \frac{7}{9}$ of the normal price.

$$\frac{7}{9} \times \text{Normal}_B = 245$$
$$\text{Normal}_B = 245 \times \frac{9}{7} = 35 \times 9 = 315 \text{ euros}$$

Step 3: Find the difference.
Difference = 320 - 315 = 5 euros

FINAL ANSWER
5 euros

PAST PAPER QUESTION

Q#: 9 **Marks: 3**

TOPIC: **Powers, Roots & Standard Form**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

9 (a) Write 5.76×10^4 as an ordinary number.

....
(1)

(b) Work out $\dfrac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

....
(2)

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Powers, Roots & Standard Form**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Move the decimal point 4 places to the right:

$$5.76 \times 10^4 = 57600$$

FINAL ANSWER
57600

Step 1: Expand the numerator.

$$3 \times 10^5 = 300,000$$
$$8 \times 10^3 = 8,000$$
$$Numerator = 300,000 + 8,000 = 308,000 = 3.08 \times 10^5$$

Step 2: Perform the division.

$$\frac{3.08 \times 10^5}{4 \times 10^{-2}} = \left(\frac{3.08}{4}\right) \times 10^{5-(-2)}$$
$$= 0.77 \times 10^7$$

Step 3: Convert to strict standard form ($1 \leq A < 10$).

$$0.77 \times 10^7 = 7.7 \times 10^6$$

FINAL ANSWER
7.7×10^6

PAST PAPER QUESTION Q#: 10 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

10 There are 5 potatoes in a bag.
The mean weight of the 5 potatoes is 217 grams.

One more potato, of weight 175 grams, is put into the bag.

Work out the mean weight of the 6 potatoes in the bag.

.... grams

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 3

TOPIC: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Calculate the total weight of the initial 5 potatoes.

$$Total_5 = Mean \times Number = 217 \times 5 = 1085 grams$$

Step 2: Calculate the new total weight.

$$Total_6 = 1085 + 175 = 1260 grams$$

Step 3: Calculate the new mean.

$$\text{New Mean} = \frac{1260}{6} = 210 \text{ grams}$$

FINAL ANSWER

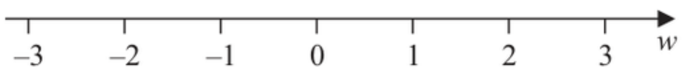
210

PAST PAPER QUESTION Q#: 11 Marks: 4

TOPIC: **Graphing Inequalities**

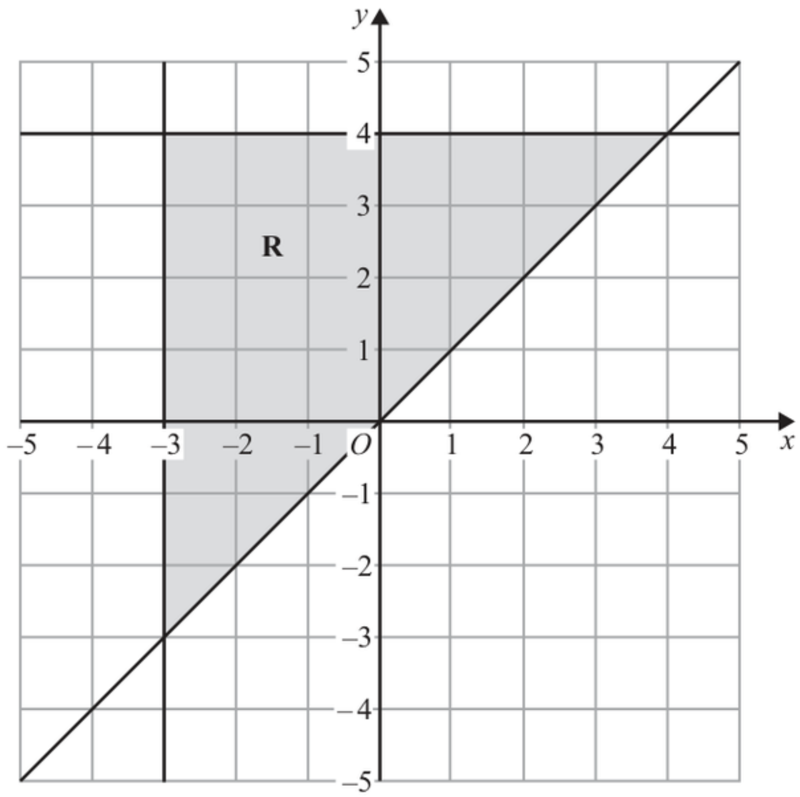
Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

11 (a) On the number line, represent the inequality $w < 1$



(1)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

....

(3)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 4

TOPIC: **Graphing Inequalities**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Draw an open circle at 1 and an arrow pointing to the left.
Step 1: Identify the vertical boundary. The region is to the right of the vertical line at $x =$ - 3. Inequality 1: $x \geq$ - 3
Step 2: Identify the horizontal boundary. The region is below the horizontal line at $y = 4$. Inequality 2: $y \leq 4$
Step 3: Identify the diagonal boundary. The line passes through (0,0), (1,1), (-1,-1), etc., meaning its equation is $y = x$. The shaded region is above this line. Inequality 3: $y \geq x$

FINAL ANSWER
$x \geq$ - 3, $y \leq 4$, $y \geq x$

Always pick a test point inside the shaded region, like (0,2), to verify your inequalities. For $y \geq x$, $2 \geq 0$ is true!

PAST PAPER QUESTION Q#: 12 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

12 The value of a house

- decreased by 5% from the end of 2021 to the end of 2022
- increased by 20% from the end of 2022 to the end of 2023
- increased by 6.5% from the end of 2023 to the end of 2024

Calculate the percentage by which the value of the house increased from the end of 2021 to the end of 2024

.... %

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Write down the multiplier for each year.
Decrease by 5% $\implies$ Multiplier = 1 - 0.05 = 0.95
Increase by 20% $\implies$ Multiplier = 1 + 0.20 = 1.20
Increase by 6.5% $\implies$ Multiplier = 1 + 0.065 = 1.065
Step 2: Calculate the combined overall multiplier.

$$OverallMultiplier = 0.95 \times 1.20 \times 1.065 = 1.2141$$

Step 3: Convert the overall multiplier back to a percentage increase.
1.2141 - 1 = 0.2141 = 21.41%

FINAL ANSWER
21.41%

PAST PAPER QUESTION Q#: 13 Marks: 2

Topic: **Powers, Roots & Standard Form**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

13 Work out $\left(9 \times 10^{80}\right)^3$

Give your answer in standard form.

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Step 1: Apply the power of 3 to both parts.

$$(9 \times 10^{80})^3 = 9^3 \times (10^{80})^3$$

Step 2: Evaluate both components.

$$9^3 = 729$$
$$(10^{80})^3 = 10^{80\times3} = 10^{240}$$

Step 3: Combine and adjust to standard form.

$$729 \times 10^{240} = 7.29 \times 10^2 \times 10^{240} = 7.29 \times 10^{242}$$

FINAL ANSWER

7.29×10^{242}

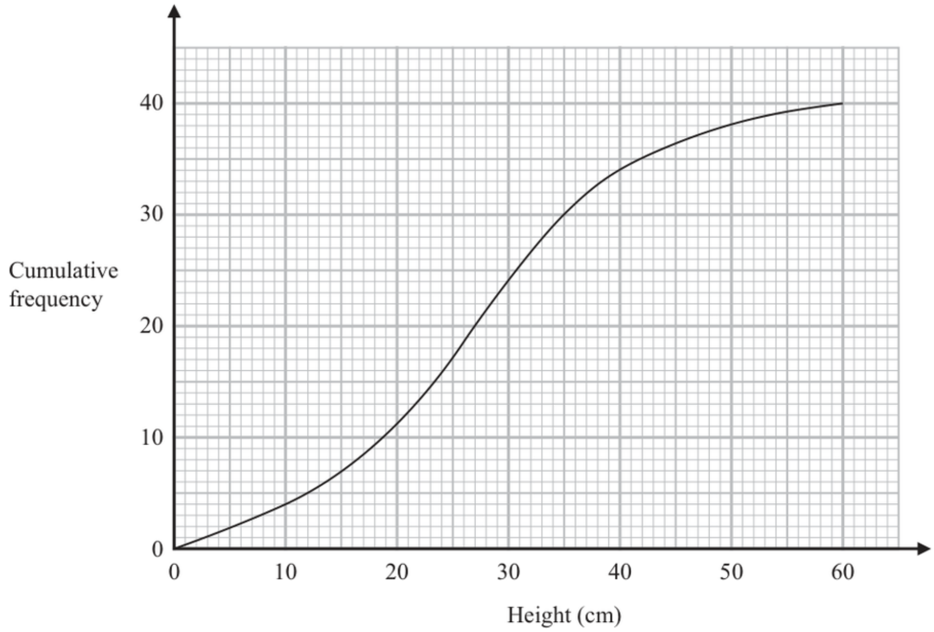
PAST PAPER QUESTION

Q#: 14 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



(a) Use the graph to find an estimate for the median height.

.... cm

(1)

(b) Use the graph to find an estimate for the interquartile range of the heights.

.... cm

(2)

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

.... euros

(2)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION Q#: 14 Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Part (a) Median: Total plants = 40. The median position is at cumulative frequency = 40/2 = 20. Reading across from CF = 20 to the curve and down to the x-axis gives:

FINAL ANSWER
a 30 cm

Part (b) Interquartile Range (IQR):
Lower Quartile (Q1) is at CF = 10. Reading the graph $\Longrightarrow$ Height = 20 cm.
Upper Quartile (Q3) is at CF = 30. Reading the graph $\Longrightarrow$ Height = 40 cm.
IQR = $Q3$ - $Q1$ = 40 - 20 = 20 cm

FINAL ANSWER
b 20 cm

Part (c) Premium plants revenue: Premium plants have height > 40 cm. From the graph, cumulative frequency at 40 cm is 30. This means 30 plants are $\leq$ 40 cm.
Number of premium plants = 40 - 30 = 10 plants

$$TotalMoney = 10plants \times 30euros = 300euros$$

FINAL ANSWER
c 300 euros

PAST PAPER QUESTION

Q#: 15 **Marks: 4**

TOPIC: **Circle Theorems**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

15 *A*, *B*, *C* and *D* are points on a circle with centre *O*

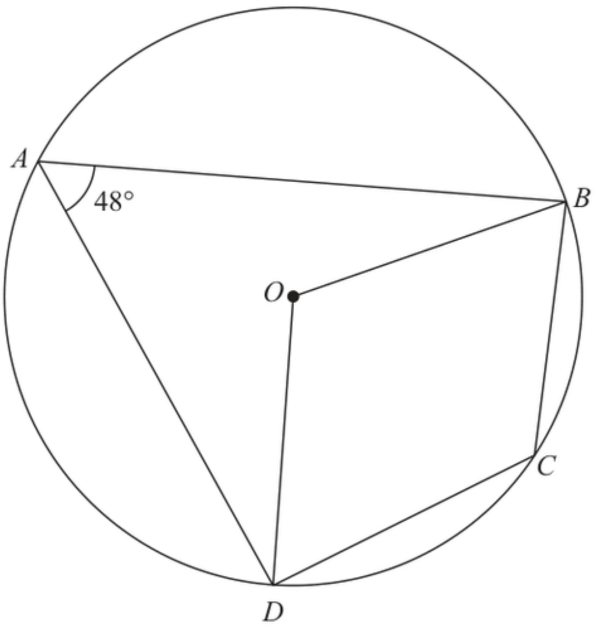

Diagram **NOT** accurately drawn

Angle $DAB = 48°$

(a) (i) Work out the size of the obtuse angle *DOB*

.... °

(1)

(ii) Give a reason for your answer.

....

(1)

(b) (i) Work out the size of angle *BCD*

.... °

(1)

(ii) Give a reason for your answer.

....

(1)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION **Q#: 15** **Marks: 4**

TOPIC: **Circle Theorems**

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WORKED SOLUTION

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Part (a)(i) Angle *DOB*:
$\angle DOB = 2 \times 48° = 96°$

FINAL ANSWER
(a)(i) 96°

Part (a)(ii) Reason: The angle subtended by an arc at the centre is twice the angle subtended at the circumference.

Part (b)(i) Angle *BCD*:
$\angle BCD = 180°$ - $48° = 132°$

FINAL ANSWER
(b)(i) 132°

Part (b)(ii) Reason: Opposite angles in a cyclic quadrilateral sum to 180°.

PAST PAPER QUESTION

Q#: 16 **Marks: 2**

TOPIC: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

16 11 people were asked how long, in minutes, they stayed in a coffee shop last Friday.

Here are the results.

14 15 16 18 19 20 22 25 28 40 50

Work out the interquartile range of these times.

.... minutes

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Statistics Toolkit**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

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Data (sorted): 14, 15, 16, 18, 19, 20, 22, 25, 28, 40, 50.

$$Number of data points n = 11.$$

Step 1: Find the Median (Q2). The median is the $\left(\frac{11+1}{2}\right) = 6^{\text{th}}$ value.

$$Median = 20$$

Step 2: Find Lower Quartile (Q1) and Upper Quartile (Q3). Q1 is the median of the lower 5 values (14, 15, **16**, 18, 19).

$$Q1 = 16$$

Q3 is the median of the upper 5 values (22, 25, **28**, 40, 50).

$$Q3 = 28$$

Step 3: Calculate the Interquartile Range (IQR).
IQR = $Q3$ - $Q1$ = 28 - 16 = 12

FINAL ANSWER
12 minutes

PAST PAPER QUESTION Q#: 17 Marks: 3

TOPIC: **Direct & Inverse Proportion**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

17 T is inversely proportional to $\sqrt{m}$

$T = 15$ when $m = 36$

Find a formula for T in terms of m

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Direct & Inverse Proportion**

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WORKED SOLUTION

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Step 1: Set up the proportionality equation.

$$T = \frac{k}{\sqrt{m}}$$

Step 2: Substitute the given values to find constant *k*.

$$15 = \frac{k}{\sqrt{36}}$$

$$15 = \frac{k}{6}$$

$$k = 15 \times 6 = 90$$

Step 3: Write the final formula.

$$T = \frac{90}{\sqrt{m}}$$

FINAL ANSWER

$T = \frac{90}{\sqrt{m}}$

PAST PAPER QUESTION

Q#: 18 **Marks: 4**

TOPIC: **Rearranging Formulas**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

18 Make w the subject of the formula $p = \sqrt{\dfrac{7w + y}{cw + k}}$

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Rearranging Formulas**

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WORKED SOLUTION

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Step 1: Square both sides to remove the square root.

$$p^2 = \frac{7w + y}{cw + k}$$

Step 2: Multiply both sides by the denominator ($cw+k$).

$$p^2(cw + k) = 7w + y$$

Step 3: Expand the bracket.

$$p^2cw + p^2k = 7w + y$$

Step 4: Collect all terms involving w on one side.
p^2cw - $7w = y$ - p^2k
Step 5: Factor out w.
$w(p^2c$-$7) = y$ - p^2k
Step 6: Divide to isolate w.

$$w = \frac{y - p^2k}{p^2c - 7}$$

FINAL ANSWER
$w = \frac{y-p^2k}{p^2c-7}$

PAST PAPER QUESTION Q#: 19 Marks: 2

TOPIC: **Volume & Surface Area**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

19 A sphere has a diameter of 24 cm

Find the surface area of the sphere.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 19 is 2 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 2

TOPIC: **Volume & Surface Area**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Find the radius.

$$r = \frac{24}{2} = 12 \text{ cm}$$

Step 2: Apply the surface area formula.
Surface Area $= 4\pi r^2$
$= 4 \times \pi \times 12^2$
$= 4 \times \pi \times 144$
$= 576\pi \approx 1809.557... \text{ cm}^2$
Step 3: Round to 3 significant figures. $1809.557 \to 1810$ (the 1, 8, and 1 are the first three sig figs).

FINAL ANSWER
1810 cm^2

PAST PAPER QUESTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x^2 + y^2 = 3xy - 11$$
$$y = x + 2$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Substitute $y = (x+2)$ into the quadratic equation.
$x^2 + (x+2)^2 = 3x(x+2) - 11$
Step 2: Expand all brackets.
$x^2 + (x^2+4x+4) = 3x^2 + 6x - 11$
$2x^2 + 4x + 4 = 3x^2 + 6x - 11$
Step 3: Move all terms to one side to set the equation to zero.
$0 = 3x^2 - 2x^2 + 6x - 4x - 11 - 4$
$x^2 + 2x - 15 = 0$
Step 4: Factorise the quadratic equation.
$(x+5)(x-3) = 0$
So, $x = -5$ or $x = 3$.
Step 5: Find corresponding y values using $y = x + 2$.
If $x = -5 \implies y = -5 + 2 = -3$
If $x = 3 \implies y = 3 + 2 = 5$

FINAL ANSWER
(-5,-3) and (3,5)

PAST PAPER QUESTION Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

20 Solve the simultaneous equations

$$x^2 + y^2 = 3xy - 11$$
$$y = x + 2$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 5

TOPIC: **Simultaneous Equations**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Substitute $y = (x+2)$ into the quadratic equation.
$x^2 + (x+2)^2 = 3x(x+2) - 11$
Step 2: Expand all brackets.
$x^2 + (x^2+4x+4) = 3x^2 + 6x - 11$
$2x^2 + 4x + 4 = 3x^2 + 6x - 11$
Step 3: Move all terms to one side to set the equation to zero.
$0 = 3x^2 - 2x^2 + 6x - 4x - 11 - 4$
$x^2 + 2x - 15 = 0$
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$(x+5)(x-3) = 0$
So, $x = -5$ or $x = 3$.
Step 5: Find corresponding y values using $y = x + 2$.
If $x = -5 \implies y = -5 + 2 = -3$
If $x = 3 \implies y = 3 + 2 = 5$

FINAL ANSWER
(-5,-3) and (3,5)

PAST PAPER QUESTION Q#: 21 Marks: 7

TOPIC: **Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

21 The functions f and g are defined as

$$\mathrm{f}(x) = 5x - 3$$

$$\mathrm{g}(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$$

(a) Find g(−1)

....

(1)

(b) Find fg(x)

Give your answer in its simplest form.

fg(x) = ...

(2)

$\mathrm{g}(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$

(c) Express the inverse function g^{-1} in the form $\mathrm{g}^{-1}(x) = \ldots$

$\mathrm{g}^{-1}(x)$ = ...

(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

TOPIC: **Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

(a) Find g(-1):
g(-1) = 4(-1)^2 + 16(-1) - 9 = 4 - 16 - 9 = - 21

FINAL ANSWER
a - 21

(b) Find $fg(x)$:
$fg(x) = f(g(x))$ = 5(4x^2+16x-9) - 3
= 20x^2 + 80x - 45 - 3 = 20x^2 + 80x - 48

FINAL ANSWER
b 20x^2 + 80x - 48

Let y = 4x^2 + 16x - 9. We must rearrange to make x the subject.
Step 1: Complete the square for the x terms.
y = 4(x^2+4x) - 9
y = 4((x+2)^2-4) - 9
y = 4(x+2)^2 - 16 - 9
y = 4(x+2)^2 - 25
Step 2: Rearrange for x.

$$y + 25 = 4(x + 2)^2$$
$$\frac{y + 25}{4} = (x + 2)^2$$
$$x + 2 = \pm\sqrt{\frac{y + 25}{4}}$$

Since $x \geq$ - 2, x + 2 is positive, so we take the positive root.

$$x + 2 = \frac{\sqrt{y + 25}}{2}$$
$$x = \frac{\sqrt{y + 25}}{2} - 2$$

Step 3: Swap y for x to write the inverse function.

FINAL ANSWER
$g^{-1}(x) = \frac{\sqrt{x+25}}{2} - 2$

When finding the inverse of a quadratic, always complete the square. The restricted domain ($x \geq$ - 2) tells you whether to take the positive or negative square root!

PAST PAPER QUESTION Q#: 21 Marks: 7

TOPIC: **Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

21 The functions f and g are defined as

$$f(x) = 5x - 3$$

$$g(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$$

(a) Find g(−1)

....

(1)

(b) Find fg(x)

Give your answer in its simplest form.

fg(x) = ...

(2)

$$g(x) = 4x^2 + 16x - 9 \qquad \text{where} \quad x \geqslant -2$$

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \ldots$

$g^{-1}(x)$ = ...

(4)

(Total for Question 21 is 7 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 7

TOPIC: **Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

(a) Find g(-1):
g(-1) = 4(-1)^2 + 16(-1) - 9 = 4 - 16 - 9 = - 21

FINAL ANSWER
a - 21

(b) Find $fg(x)$:
$fg(x) = f(g(x))$ = 5(4x^2+16x-9) - 3
= 20x^2 + 80x - 45 - 3 = 20x^2 + 80x - 48

FINAL ANSWER
b 20x^2 + 80x - 48

Let y = 4x^2 + 16x - 9. We must rearrange to make x the subject.

Step 1: Complete the square for the x terms.
y = 4(x^2+4x) - 9
y = 4((x+2)^2-4) - 9
y = 4(x+2)^2 - 16 - 9
y = 4(x+2)^2 - 25

Step 2: Rearrange for x.

$$y + 25 = 4(x + 2)^2$$
$$\frac{y + 25}{4} = (x + 2)^2$$
$$x + 2 = \pm\sqrt{\frac{y + 25}{4}}$$

Since $x \geq$ - 2, x + 2 is positive, so we take the positive root.

$$x + 2 = \frac{\sqrt{y + 25}}{2}$$
$$x = \frac{\sqrt{y + 25}}{2} - 2$$

Step 3: Swap y for x to write the inverse function.

FINAL ANSWER
$g^{-1}(x) = \frac{\sqrt{x+25}}{2} - 2$

When finding the inverse of a quadratic, always complete the square. The restricted domain ($x \geq$ - 2) tells you whether to take the positive or negative square root!

PAST PAPER QUESTION Q#: 22 Marks: 6

TOPIC: **Differentiation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11
The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b
Show clear algebraic working.

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION Q#: 22 Marks: 6

TOPIC: **Differentiation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Differentiate the curve to find the gradient expression.

$$\frac{dy}{dx} = 3x^2 - 16$$

Step 2: Set the gradient to 11 to find the x-coordinates.

$3x^2$ - 16 = 11

$$3x^2 = 27$$

$x^2 = 9 \implies x = 3$ or $x = -3$

Step 3: Find the corresponding y-coordinates on the curve.

For $x = 3$: $y = (3)^3 - 16(3) + 7 = 27 - 48 + 7 = -14$

Point 1: (3,-14)

For $x = -3$: $y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$

Point 2: (-3,28)

Step 4: Find the y-intercept (b) for each tangent ($y = 11x + b$).

Using (3,-14):

$-14 = 11(3) + b \implies -14 = 33 + b \implies b = -47$

Using (-3,28):

$28 = 11(-3) + b \implies 28 = -33 + b \implies b = 61$

FINAL ANSWER

$b = -47$ and 61

PAST PAPER QUESTION **Q#: 22** **Marks: 6**

TOPIC: **Differentiation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11
The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b
Show clear algebraic working.

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 6

TOPIC: **Differentiation**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Differentiate the curve to find the gradient expression.

$$\frac{dy}{dx} = 3x^2 - 16$$

Step 2: Set the gradient to 11 to find the x-coordinates.

$3x^2 - 16 = 11$

$$3x^2 = 27$$

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Step 3: Find the corresponding y-coordinates on the curve.

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Point 1: (3,-14)

For $x = -3$: $y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$

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Step 4: Find the y-intercept (b) for each tangent ($y = 11x + b$).

Using (3,-14):

$-14 = 11(3) + b \implies -14 = 33 + b \implies b = -47$

Using (-3,28):

$28 = 11(-3) + b \implies 28 = -33 + b \implies b = 61$

FINAL ANSWER

$b = -47$ and 61

PAST PAPER QUESTION

Q#: 23 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm^2)	**Volume (cm^3)**
Shape **P**	200	672
Shape **Q**	450	

Work out the volume of shape **Q**

.... cm^3

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Find the Area Scale Factor (ASF).

$$\text{ASF} = \frac{\text{Area of Q}}{\text{Area of P}} = \frac{450}{200} = \frac{9}{4} = 2.25$$

Step 2: Find the Length Scale Factor (LSF).

$$\text{LSF} = \sqrt{\text{ASF}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

Step 3: Find the Volume Scale Factor (VSF).

$$\text{VSF} = (\text{LSF})^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8} = 3.375$$

Step 4: Calculate the volume of Shape Q.

$$\text{Volume of Q} = \text{Volume of P} \times \text{VSF} = 672 \times \frac{27}{8}$$
$$= 84 \times 27 = 2268$$

FINAL ANSWER

$2268\ \text{cm}^3$

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

23 Shape **P** is similar to shape **Q**

The table shows some information about shape **P** and shape **Q**

	Surface area (cm^2)	**Volume (cm^3)**
Shape **P**	200	672
Shape **Q**	450	

Work out the volume of shape **Q**

.... cm^3

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Find the Area Scale Factor (ASF).

$$\text{ASF} = \frac{\text{Area of Q}}{\text{Area of P}} = \frac{450}{200} = \frac{9}{4} = 2.25$$

Step 2: Find the Length Scale Factor (LSF).

$$\text{LSF} = \sqrt{\text{ASF}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

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Step 4: Calculate the volume of Shape Q.

$$\text{Volume of Q} = \text{Volume of P} \times \text{VSF} = 672 \times \frac{27}{8}$$

$$= 84 \times 27 = 2268$$

FINAL ANSWER

2268 cm^3

PAST PAPER QUESTION Q#: 24 Marks: 6

TOPIC: **Vectors**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

24 The diagram shows a quadrilateral $OABC$

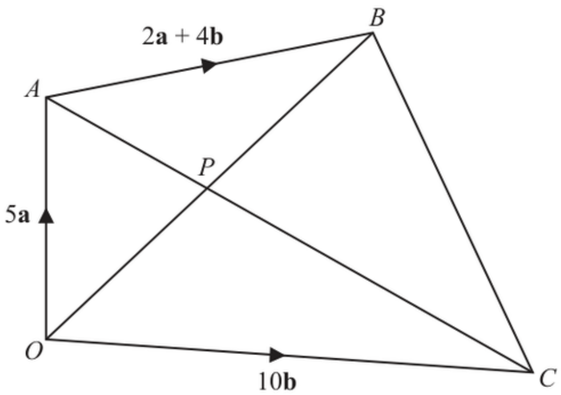


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 5\mathbf{a}$ $\quad\overrightarrow{OC} = 10\mathbf{b}$ $\quad\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**

$\overrightarrow{AC} =$ ....

(1)

(b) Find $\overrightarrow{OB}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{OB} =$ ....

(1)

The point P is such that APC and OPB are straight lines.

(c) Using a vector method, find $\overrightarrow{OP}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OP} =$ ....

(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

(a) Find $\overrightarrow{AC}$:

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\overrightarrow{OA} + \overrightarrow{OC} = -5\mathbf{a} + 10\mathbf{b}$$

FINAL ANSWER

a - 5**a** + 10**b**

(b) Find $\overrightarrow{OB}$:

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{a} + (2\mathbf{a} + 4\mathbf{b}) = 7\mathbf{a} + 4\mathbf{b}$$

FINAL ANSWER

b 7**a** + 4**b**

Step 1: Set up vector equations for $\overrightarrow{OP}$. **Since *OPB* is a straight line, $\overrightarrow{OP}$ is a scalar multiple of $\overrightarrow{OB}$. Let this be λ.

$$\overrightarrow{OP} = \lambda(7\mathbf{a} + 4\mathbf{b}) = 7\lambda\mathbf{a} + 4\lambda\mathbf{b}$$

Since P lies on the straight line AC, $\overrightarrow{AP} = \mu\overrightarrow{AC}$ for some scalar μ.

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 5\mathbf{a} + \mu(-5\mathbf{a} + 10\mathbf{b})$$

$$\overrightarrow{OP} = (5 - 5\mu)\mathbf{a} + 10\mu\mathbf{b}$$

Step 2: Equate the coefficients of a and b.

$7\lambda = 5$ - $5\mu - (1)$

$$4\lambda = 10\mu \implies \mu = \frac{4}{10}\lambda = 0.4\lambda \quad — (2)$$

Step 3: Solve for *λ*. Substitute (2) into (1):

$7\lambda = 5$ - $5(0.4\lambda)$

$7\lambda = 5$ - 2λ

$$9\lambda = 5 \implies \lambda = \frac{5}{9}$$

Step 4: Find $\overrightarrow{OP}$.

$$\overrightarrow{OP} = \frac{5}{9}(7\mathbf{a} + 4\mathbf{b}) = \frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

FINAL ANSWER

$\frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$

PAST PAPER QUESTION

Q#: 24 **Marks: 6**

TOPIC: **Vectors**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

24 The diagram shows a quadrilateral *OABC*

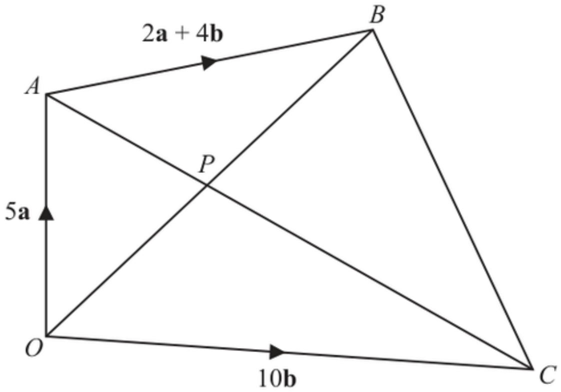


Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 5\mathbf{a}$ $\overrightarrow{OC} = 10\mathbf{b}$ $\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**

$\overrightarrow{AC} =$ ....

(1)

(b) Find $\overrightarrow{OB}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{OB} =$ ....

(1)

The point *P* is such that *APC* and *OPB* are straight lines.

(c) Using a vector method, find $\overrightarrow{OP}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OP} =$ ....

(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

(a) Find $\overrightarrow{AC}$:

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\overrightarrow{OA} + \overrightarrow{OC} = -5\mathbf{a} + 10\mathbf{b}$$

FINAL ANSWER

a - 5**a** + 10**b**

(b) Find $\overrightarrow{OB}$:

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{a} + (2\mathbf{a} + 4\mathbf{b}) = 7\mathbf{a} + 4\mathbf{b}$$

FINAL ANSWER

b 7**a** + 4**b**

**Step 1: Set up vector equations for $\overrightarrow{OP}$. **Since *OPB* is a straight line, $\overrightarrow{OP}$ is a scalar multiple of $\overrightarrow{OB}$. Let this be λ.

$$\overrightarrow{OP} = \lambda(7\mathbf{a} + 4\mathbf{b}) = 7\lambda\mathbf{a} + 4\lambda\mathbf{b}$$

Since P lies on the straight line AC, $\overrightarrow{AP} = \mu\overrightarrow{AC}$ for some scalar μ.

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 5\mathbf{a} + \mu(-5\mathbf{a} + 10\mathbf{b})$$

$$\overrightarrow{OP} = (5 - 5\mu)\mathbf{a} + 10\mu\mathbf{b}$$

Step 2: Equate the coefficients of a and b.

$7\lambda = 5$ - $5\mu - (1)$

$$4\lambda = 10\mu \implies \mu = \frac{4}{10}\lambda = 0.4\lambda \quad \text{— (2)}$$

Step 3: Solve for *λ*. Substitute (2) into (1):

$7\lambda = 5$ - $5(0.4\lambda)$

$7\lambda = 5$ - 2λ

$$9\lambda = 5 \implies \lambda = \frac{5}{9}$$

Step 4: Find $\overrightarrow{OP}$.

$$\overrightarrow{OP} = \frac{5}{9}(7\mathbf{a} + 4\mathbf{b}) = \frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

FINAL ANSWER

$\frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$

PAST PAPER QUESTION Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

25 A curve has equation $y = \mathrm{f}(x)$

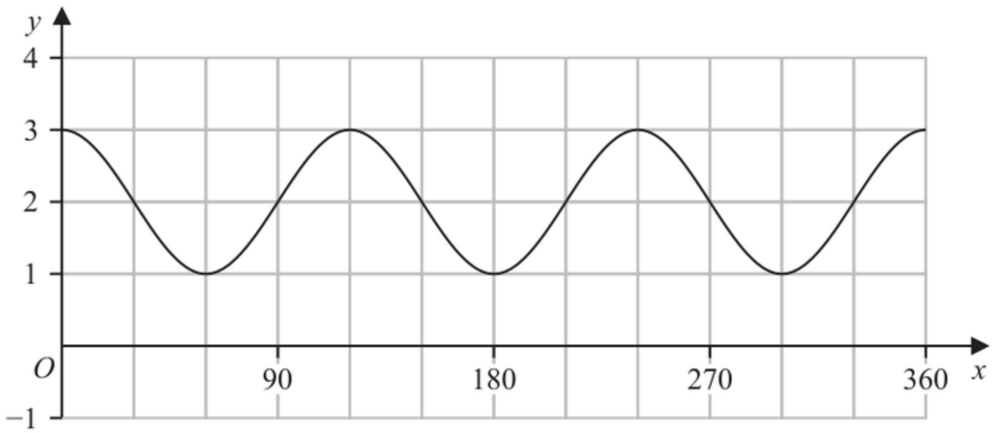
There is one minimum point on this curve.

The coordinates of this minimum point are $(8, 1)$

(a) Write down the coordinates of the minimum point on the curve with equation $y = \mathrm{f}(4x)$

(.............................. , ...)
(1)

The diagram shows the graph of $y = \cos(ax) + b$ for $0 \leqslant x \leqslant 360$



(b) (i) Find the value of a

a = ...
(1)

(ii) Find the value of b

b = ...
(1)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

The transformation $y = f(4x)$ represents a horizontal stretch by a scale factor of $\frac{1}{4}$. The x-coordinate is multiplied by $\frac{1}{4}$ while the y-coordinate stays the same.

$$\text{New minimum} = \left(8 \times \frac{1}{4}, 1\right) = (2, 1)$$

FINAL ANSWER
(2,1)

Step 1: Find the vertical shift *b*. The standard $y = \cos(x)$ graph fluctuates between - 1 and 1, centered at $y = 0$. The given graph fluctuates between a minimum of $y = 1$ and a maximum of $y = 3$. The center line is $\frac{3+1}{2} = 2$. Therefore, $b = 2$.
Step 2: Find the frequency multiplier *a*. The period of the given graph completes one full cycle between $x = 0$ and $x = 90°$ (it starts at 3, dips to 1, and returns to 3 at 90°). The standard cosine period is 360°.

$$\text{Period} = \frac{360°}{a}$$
$$90° = \frac{360°}{a} \implies a = 4$$

FINAL ANSWER
$a = 4, b = 2$

PAST PAPER QUESTION

Q#: 25 **Marks: 3**

TOPIC: **Graphs of Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

25 A curve has equation $y = \mathrm{f}(x)$

There is one minimum point on this curve.

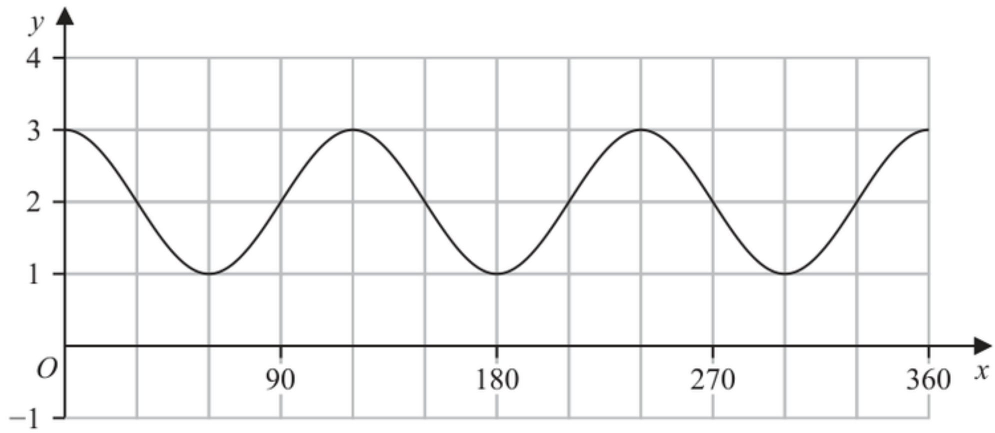
The coordinates of this minimum point are $(8, 1)$

(a) Write down the coordinates of the minimum point on the curve with equation $y = \mathrm{f}(4x)$

(.............................. , ...)

(1)

The diagram shows the graph of $y = \cos(ax) + b$ for $0 \leqslant x \leqslant 360$



(b) (i) Find the value of a

a = ...

(1)

(ii) Find the value of b

b = ...

(1)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

The transformation $y = f(4x)$ represents a horizontal stretch by a scale factor of $\frac{1}{4}$. The x-coordinate is multiplied by $\frac{1}{4}$ while the y-coordinate stays the same.

$$\text{New minimum} = \left(8 \times \frac{1}{4}, 1\right) = (2, 1)$$

FINAL ANSWER
(2,1)

Step 1: Find the vertical shift *b*. The standard $y = \cos(x)$ graph fluctuates between - 1 and 1, centered at $y = 0$. The given graph fluctuates between a minimum of $y = 1$ and a maximum of $y = 3$. The center line is $\frac{3+1}{2} = 2$. Therefore, $b = 2$.
Step 2: Find the frequency multiplier *a*. The period of the given graph completes one full cycle between $x = 0$ and $x = 90°$ (it starts at 3, dips to 1, and returns to 3 at 90°). The standard cosine period is 360°.

$$\text{Period} = \frac{360°}{a}$$
$$90° = \frac{360°}{a} \implies a = 4$$

FINAL ANSWER
$a = 4, b = 2$

PAST PAPER QUESTION

Q#: 26 **Marks: 5**

TOPIC: **Sequences**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

26 The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

$N =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION Q#: 26 Marks: 5

TOPIC: **Sequences**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Find the common difference *d*. The n^{th} term is given by $u\_\{n\} = a + (n\text{-}1)d$.

$$u_{20} = 10 + 19d = 86$$
$$19d = 76$$
$$d = \frac{76}{19} = 4$$

Step 2: Use the sum formula to form an equation in terms of *N*.

$$S_N = \frac{N}{2}\left[2a + (N-1)d\right]$$
$$Substitutetheknownvalues S_N = 5194, a = 10, andd = 4:$$
$$5194 = \frac{N}{2}\left[2(10) + (N-1)(4)\right]$$

Step 3: Simplify and solve the quadratic equation. Multiply by 2 to clear the fraction: $10388 = N\backslash[20+4N\text{-}4\backslash]$

$$10388 = N(4N + 16)$$

$4N^2 + 16N$ - $10388 = 0$
Divide the entire equation by 4 to make it simpler:
$N^2 + 4N$ - $2597 = 0$

Step 4: Use the quadratic formula to find *N*.

$$N = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$N = \frac{-4 \pm \sqrt{16 - 4(1)(-2597)}}{2}$$
$$N = \frac{-4 \pm \sqrt{16 + 10388}}{2}$$
$$N = \frac{-4 \pm \sqrt{10404}}{2}$$
$$N = \frac{-4 \pm 102}{2}$$

Since N must be a positive integer, we discard the negative root:

$$N = \frac{-4 + 102}{2} = \frac{98}{2} = 49$$

FINAL ANSWER
$N = 49$

Always double-check that your N (number of terms) evaluates to a positive whole integer. If it ends up as a decimal or fraction, recheck your arithmetic progression formulas.

PAST PAPER QUESTION **Q#: 26** **Marks: 5**

TOPIC: **Sequences**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

26 The first term of an arithmetic series is 10
The 20th term of the series is 86

The sum of the first N terms of the series is 5194

Work out the value of N
Show your working clearly.

$N =$ ....

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 5

TOPIC: **Sequences**

Nov 2025 4WM2H - Nov2025 | 4WM2H | Unit 2

WORKED SOLUTION

Dr. Eslam Ahmed

Step 1: Find the common difference *d*. The n^{th} term is given by $u\_\{n\} = a + (n\text{-}1)d$.

$$u_{20} = 10 + 19d = 86$$
$$19d = 76$$
$$d = \frac{76}{19} = 4$$

Step 2: Use the sum formula to form an equation in terms of *N*.

$$S_N = \frac{N}{2}\left[2a + (N-1)d\right]$$
$$Substitutetheknownvalues S_N = 5194, a = 10, andd = 4:$$
$$5194 = \frac{N}{2}\left[2(10) + (N-1)(4)\right]$$

Step 3: Simplify and solve the quadratic equation. Multiply by 2 to clear the fraction: $10388 = N\backslash[20+4N\text{-}4\backslash]$

$$10388 = N(4N + 16)$$

$4N^2 + 16N$ - $10388 = 0$
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$$N = \frac{-4 \pm \sqrt{10404}}{2}$$
$$N = \frac{-4 \pm 102}{2}$$

Since N must be a positive integer, we discard the negative root:

$$N = \frac{-4 + 102}{2} = \frac{98}{2} = 49$$

FINAL ANSWER

$N = 49$

Always double-check that your N (number of terms) evaluates to a positive whole integer. If it ends up as a decimal or fraction, recheck your arithmetic progression formulas.